

KEOKUK, IA, USA

WMO: 725456

Lat: 40.460N

Lon: 91.428W

Elev: 205

StdP: 98.89

Time Zone: -6.00 (NAC)

Period: 95-19

WBAN: 04921

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-17.5	-14.8	-22.4	0.5	-16.1	-20.1	0.6	-13.3	11.5	-2.4	10.4	-1.3	3.0	320	0.481

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.0	34.0	25.3	32.6	24.9	31.3	24.1	26.9	32.2	26.0	31.3	25.2	30.1	3.9	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.2	20.9	30.2	24.2	19.6	29.4	23.1	18.3	28.5	85.6	31.5	82.0	31.2	77.6	30.3	29.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.7	8.4	7.5	DB	-22.1	36.1	3.7	1.8	-24.7	37.4	-26.9	38.4	-29.0	39.4	-31.7	40.7	(4)
(5)			WB	-22.1	28.1	3.6	1.0	-24.6	28.8	-26.7	29.4	-28.7	29.9	-31.3	30.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	12.0	-2.7	-0.7	5.8	12.1	18.0	22.8	25.1	24.1	20.0	13.1	6.2	0.0		
	DBStd	11.01	6.78	6.53	6.47	5.39	4.58	3.28	3.26	3.00	4.50	5.28	5.82	5.85		
	HDD10.0	1387	394	304	163	38	2	0	0	0	1	28	142	314		
	HDD18.3	2986	652	534	392	196	64	4	1	33	176	365	569			
	CDD10.0	2131	1	4	32	102	249	386	467	436	299	126	28	2		
	CDD18.3	691	0	0	2	10	53	139	209	178	82	14	1	0		
	CDH23.3	6457	0	0	14	102	460	1276	2181	1589	716	116	3	0		
	CDH26.7	2369	0	0	1	17	121	459	922	590	236	24	0	0		
(14) Wind	WSAvg	3.4	3.8	3.8	4.1	4.2	3.4	2.9	2.5	2.4	2.7	3.3	3.7	3.6		
(15) Precipitation	PrecAvg	937	35	29	70	91	114	98	116	82	115	82	63	45		
	PrecMax	1425	218	76	188	204	265	284	309	254	323	210	209	145		
	PrecMin	463	1	3	11	17	16	4	16	34	0	3	7	3		
	PrecStd	229	41	16	40	48	68	61	75	51	79	58	44	32		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	16.4	19.1	25.8	28.8	32.3	33.8	37.4	36.1	33.9	29.8	22.9	17.8		
		MCWB	11.9	12.7	17.7	19.2	23.7	24.8	25.0	23.6	24.0	22.5	14.9	14.1		
	2%	DB	12.4	15.0	22.2	26.2	29.5	32.5	34.8	33.7	32.0	26.3	20.2	13.0		
		MCWB	8.9	10.4	15.9	17.1	22.3	24.5	26.2	25.2	23.5	18.5	14.6	9.6		
	5%	DB	9.1	11.3	18.8	23.8	27.6	31.2	32.8	32.1	29.7	23.9	17.6	10.5		
		MCWB	6.2	8.1	13.5	16.2	20.4	24.0	25.5	24.9	23.2	17.7	12.9	7.7		
	10%	DB	6.3	7.9	16.0	21.4	26.1	29.4	31.3	30.0	27.4	22.1	15.1	7.8		
		MCWB	3.4	5.1	11.0	14.8	19.4	23.1	24.6	24.0	21.0	16.8	10.5	5.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	13.6	14.7	19.2	20.6	25.1	26.9	28.6	28.1	25.7	22.9	17.4	15.3		
		MCDB	16.1	17.8	24.2	26.4	30.5	31.8	34.1	33.4	31.2	28.1	20.9	18.0		
	2%	WB	9.3	10.8	16.6	18.7	23.1	25.7	27.2	26.7	24.6	20.9	15.4	10.7		
		MCDB	11.8	14.3	20.4	24.5	27.9	30.9	32.4	31.9	29.8	24.4	19.0	12.5		
	5%	WB	6.0	7.9	14.3	17.2	21.8	24.8	26.3	25.7	23.4	19.1	13.4	7.9		
		MCDB	9.1	11.0	18.5	22.4	26.7	29.8	32.0	30.5	27.8	22.7	17.5	9.8		
	10%	WB	3.8	5.3	11.3	15.5	20.3	23.7	25.4	24.7	22.3	17.2	10.9	5.5		
		MCDB	6.3	8.2	15.2	20.4	24.7	28.4	30.4	28.9	26.6	21.1	14.8	7.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	8.7	8.8	10.0	10.9	10.4	10.2	10.0	9.9	11.1	10.5	9.4	7.9		
		MCDBR	12.5	12.8	13.7	14.0	11.8	11.4	11.0	11.8	11.8	12.7	12.6	11.0		
	5% WB	MCWBR	10.0	9.3	8.5	8.1	6.7	6.1	5.6	6.1	5.6	7.1	8.6	8.9		
		MCWBR	11.7	12.0	12.2	12.2	10.7	10.3	10.1	10.0	10.2	10.3	11.6	10.3		
(40) Clear-Sky Solar Irradiance	taub		0.292	0.310	0.339	0.392	0.425	0.427	0.446	0.437	0.396	0.351	0.303	0.289		
		taud	2.454	2.399	2.405	2.261	2.232	2.280	2.204	2.249	2.312	2.443	2.516	2.477		
	Ebn at Noon		875	906	914	881	856	852	831	828	840	848	857	850		
		Edh at Noon	78	95	106	131	138	132	141	132	116	91	73	71		
(44) All-Sky Solar Radiation	RadAvg		1.86	2.80	3.69	4.79	5.58	6.26	6.24	5.49	4.73	3.17	2.14	1.59		
	RadStd		0.22	0.23	0.25	0.37	0.40	0.59	0.39	0.29	0.34	0.35	0.17	0.20		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (35 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page